

Water + cation hydrolysis at the OHP

What the model does, what we found, what to ask

Pre-meeting brief, K₂SO₄ / ORR / CMK-3 forward solver

2026-05-21

Abstract

This is a concise pre-meeting brief on the water-ionization (Phase 6 α) and cation-hydrolysis (Phase 6 β) work that the PNP+BV forward solver has been chewing on for the past two weeks. Section 1 states the gap. Section 2 explains the physical mechanism the group's own framing points to. Section 3 writes down the equations actually inside the solver. Section 4 summarises each sub-step. Section 5 reports the Phase D non-identifiability verdict. Section 6 lists the gaps we've found. Section 7 is the question list for the meeting.

1 The gap, in one sentence

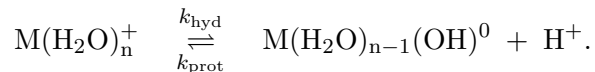
At K₂SO₄, pH 4–6, $L_{\text{eff}} \approx 100 \mu\text{m}$, the production solver matches the deck's *total* disk current to within Levich order-of-magnitude, but its partial H₂O₂ current is **flat with** V_{RHE} (a plateau set by the H⁺ diffusion Levich limit at $\approx -0.09 \text{ mA/cm}^2$ at pH 4), whereas the deck data shows a **cathodic peak** near $V_{\text{RHE}} \approx 0 \text{ V}$ that then decays. The model max H₂O₂% is 66.6 pp uniformly; the deck K₂SO₄/pH 4 mean is 50.95 pp — a uniform +15.6 pp overshoot. The whole Phase 6 investigation is about **what missing physics builds that peak**.

2 The chemistry, plain language

There are two mechanisms in the picture. Both produce H⁺ near the cathode (where ORR consumes H⁺).

(A) Water self-ionization. $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$, $K_w = 10^{-14} \text{ M}^2$ in bulk. When the cathode consumes H⁺ faster than it can diffuse from the bulk, water dissociates locally to backfill. This is universal aqueous chemistry — it always happens.

(B) Cation hydrolysis at the outer Helmholtz plane (OHP). Hydrated alkali cations, when pinned at the OHP by a strong cathodic field, can have a water molecule in their inner hydration shell deprotonate:



The leftover **MOH**⁰ is a neutral surface adsorbate; the released H⁺ enters the diffuse layer. The cation's effective $\text{p}K_a$ *drops sharply* near a polarized cathode because the local field stabilises the

deprotonated form. In bulk, K^+ pK_a is ~ 14.5 ; Singh 2016 reports it dropping to ~ 8.5 near a Cu cathode at -1 V. Cs^+ drops to ~ 4.3 .

Why (B) is the group’s preferred hypothesis

The Seitz/Mangan group documents (Linsey 2025 ACS-CATL deck slide 27, Trienens 2025 report, Co-Zhang 2019 *Angew. Chem.*, Singh 2016 *JACS*) consistently propose **(B) cation hydrolysis** as the dominant cation-dependent buffer for K_2SO_4 ORR — not sulfate buffering, not carbonate, not water self-ionization. Phase 6α (water) was a useful sub-leading correction that retired in favour of Phase 6β (cation) once the deck framing was audited.

3 How the model represents hydrolysis

Both mechanisms live inside the forward PNP+BV stack as opt-in “modules” — default-off, opt-in by a flag in the solver-params. Production runs since May 10 have both enabled (water on; cation on at $\lambda = 1$).

3.1 (A) Water self-ionization (Phase 6α)

Instead of treating H^+ and OH^- as two independent Nernst–Planck species, we use a **proton-condition** variable

$$E(y) \equiv c_H(y) - c_{OH}(y),$$

with the fast-equilibrium closure $c_H \cdot c_{OH} = K_w^{\text{eff}}$. One PDE replaces two, and the water source/sink term cancels by construction (forward and reverse produce/consume H^+ and OH^- in 1:1 ratio, so their difference E has no source). At steady state:

$$-\nabla \cdot J_E = 0, \quad J_E = -(D_H c_H + D_{OH} c_{OH}) \nabla \mu_H.$$

This is a textbook fast-water approximation. K_w^{eff} is ramped from 0 to 10^{-14} M² over a continuation ladder. The Damköhler number $\tau_{\text{diff}}/\tau_{\text{water}}$ is $\gtrsim 10^7$ everywhere, so fast equilibrium is robust as a zeroth-order model. Caveat: *textbook* bulk water dissociation rate ~ 1.4 mol/m³/s is $\sim 20\times$ smaller than the cathodic H^+ demand at $L_{\text{eff}} = 16$ μm in the Phase 6α sweep — so the fast-water closure may over-predict H^+ replenishment.

3.2 (B) Cation hydrolysis (Phase 6β v9 / v10b)

Implemented as a boundary source on the OHP, parameterised by a global surface coverage Γ_{MOH} (mol/m²) and a finite-rate kinetic law:

$$R_{\text{net}}(\sigma_S) = \underbrace{k_{\text{hyd}} c_{M^+}(0) 10^{-\Delta pK_a(\sigma_S)} \left(1 - \frac{\Gamma}{\Gamma_{\text{max}}}\right)}_{\text{forward (Langmuir-capped)}} - \underbrace{k_{\text{prot}} c_H(0) \frac{\Gamma}{\delta_{\text{OHP}}}}_{\text{reverse}}, \quad (1)$$

$$R_{\text{des}} = k_{\text{des}} \Gamma \quad (\text{open-reservoir desorption}). \quad (2)$$

At steady state, $R_{\text{net}} = R_{\text{des}}$, which gives a closed-form $\Gamma_{\text{ss}}(\lambda)$ that the orchestrator recomputes from the converged boundary values between continuation rungs (outer Picard). At $\lambda = 1$, the H^+ residual gains $+R_{\text{net}} v_H ds$ at the electrode, and the K^+ residual gains $-R_{\text{net}} v_K ds$ (one proton produced per cation neutralised).

The Singh 2016 field-dependent pK_a (SI Eq. 4). The ΔpK_a term in eq. (1) comes straight from Singh et al. 2016 *JACS* (CO₂RR-on-Cu/Ag):

$$\Delta pK_a(\sigma_S) = +2A z r_{\text{H,El}} \left(1 - \frac{r_{M-O}^2}{r_{\text{H,El}}^2}\right) \sigma_{\text{Singh}}(\sigma_S).$$

The geometric factor $1 - r_{M-O}^2/r_{\text{H,El}}^2$ is *negative* when $r_{\text{H,El}} < r_{M-O}$ (Singh’s adsorbed-CO sandwich geometry), so $\Delta pK_a < 0$ under cathodic bias — the pK_a drops as advertised. σ_{Singh} is $|\sigma_S|$ rescaled to Singh’s counts/pm² unit with an anodic clamp: $\sigma_{\text{Singh}} = \max(0, -\sigma_S) \cdot N_A/F \cdot 10^{-24}$. $z = z_{\text{eff}}$ is the *effective* cation charge (Singh Table S1); $A = 620.32$ pm. The per-cation Table S1 row plus a **back-fitted** $r_{\text{H,El}}^{\text{Cu}}$ is hard-coded verbatim from the SI:

Cation	r_M (pm)	z_{eff}	n_{hyd}	pK_a^{bulk}	$r_{\text{H,El}}^{\text{Cu}}$ (pm)	pK_a near Cu @ -1 V
Li ⁺	69	0.864	5.2	13.6	132.00	13.16
Na ⁺	102	0.900	3.5	14.2	164.99	11.44
K ⁺	138	0.919	2.6	14.5	200.98	8.49
Rb ⁺	149	0.923	2.4	14.6	211.98	7.23
Cs ⁺	170	0.930	2.1	14.8	232.97	4.32

What does *not* happen in the model. Several choices to be explicit about, because each is a question for the meeting:

- **No Stern surface-charge correction from MOH.** Γ_{MOH} is electrically neutral; it does not enter the Stern Gauss balance. Hydrolysis couples to electrostatics *purely* via the dynamic K⁺ NP profile responding to the boundary K⁺ sink.
- **Open-reservoir desorption.** MOH⁰ that desorbs is not tracked in the bulk — no 5th NP species.
- **Cu prior used for ORR-on-carbon.** $r_{\text{H,El}}$ values are back-fitted from Singh’s Cu Table S3. The deck’s electrode is CMK-3 mesoporous carbon, not Cu. We use the Cu values as a starting prior pending carbon-specific calibration.
- **Singh’s adsorbed-CO sandwich geometry.** Singh’s derivation assumes adsorbed CO sandwiched between the hydrated cation and Cu. ORR on carbon adsorbs O₂/OOH/OH/H₂O instead; the geometric assumption that fixes $r_{\text{H,El}} < r_{M-O}$ is therefore not strictly justified for our case.
- **Sulfate buffering is out of scope.** HSO₄[−]/SO₄^{2−} treated as inert spectators; only their counterion Bikerman steric appears in the Poisson source.

3.3 Locked parameter values (production, v10b)

Parameter	Value	Source
C_S (Stern cap.)	0.20 F/m ²	Bohra–Koper–Choi consensus ($\varepsilon_S = 11.3$, $L_S = 5$ Å)
$\Gamma_{\max}$	0.047 nondim ($\approx 5.6 \times 10^{-6}$ mol/m ²)	Hard-sphere monolayer, $r = 2.3$ Å for K ⁺ packing (V10A derivation, 4-test-compat tightened)
k_{des}	1.0 nondim ($k_{\text{des}}^{\text{phys}} \sim 0.2$ /s)	<i>Engineering choice</i> , Eyring prior $\Delta G \in [0.69, 0.94]$ eV at 298 K
δ_{OHP}	0.40 nm	Bohra 2019 EES prior
$r_{\text{H,El}}^{\text{Cu}}$	per-cation table above (Singh SI back-fit)	
$K_0^{R_{4e}}/K_0^{R_{2e}}$	10^{-14} (placeholder ratio)	Tuned to match qualitative selectivity; calibration source missing
$E_{eq}^{R_{2e}}, E_{eq}^{R_{4e}}$	0.695 V, 1.23 V vs RHE	Ruggiero 2022 §1 (parallel $2e^-/4e^-$ topology)

4 What we tried, step by step

The history below is the chronology from May 9 to May 12. Each sub-step delivered a passing convergence/mass-balance test before moving on; the only thing that did not pass is the *fit* (step 10).

Step	What changed	Verdict
6α sweep	Wired water self-ionization (proton-condition residual on $E = c_H - c_{OH}$); ran 8 L_{eff} values $\times$ 13 V_{RHE} points to convergence in ~ 58 min.	Levich linearity PASS , no spurious peak PASS , surface pH < 9 FAIL (pH locks at 10.58 at the smallest L_{eff} and is L_{eff} -invariant) — 6 α retired as sub-leading.
Conjecture audit	Grepped <code>data/EChemReactorModeling-Seitz-Mangano</code> to find which 6 α /6 β choices were Claude/GPT conjectures vs. deck-grounded.	6 α was wrong baseline; K ⁺ /SO ₄ ²⁻ is the deck baseline. Pivoted all 6 β work to K ⁺ . C_S , $r_{\text{H,El}}$ priors flagged as “tunables, not Ruggiero-cited.”
v9 architecture	Critique-loop with GPT (5 rounds) replaced an algebraic-shadow attempt with: <i>dynamic K⁺ NP species</i> , global Γ scalar at the OHP, finite-rate kinetics, no Stern surface-charge correction.	Architecture approved. All four “gate” unit tests pass (build / $\lambda=0$ equilibrium / Γ source sign / finite-rate $\lambda=1$ smoke at -0.40 V).
Singh SI extracted	User retrieved <code>C02RRCationSupplementaryInformation</code> pdf via institutional access; Eq. (3) verified to recover Table S1 to rounding, Eq. (4) implemented per-cation.	Field-dependent $\text{p}K_a$ is now a physical formula, not a placeholder $\beta_M \text{sgn}(\sigma_S) \sigma_S ^p$.
v10a Langmuir cap	Added $(1 - \Gamma/\Gamma_{\max})$ vacancy factor on the forward branch; without it, $k_{\text{hyd}} \geq 10^{-3}$ saturates at $\Gamma \sim 6$ –64 monolayers (unphysical).	Closed-form Picard converges; saturation behaves smoothly.
v10a' V-sweep	Locked $V_{\text{kin}} = -0.10$ V as the diagnostic voltage: branch active ($\sigma_S < 0$), not transport-limited, not cap-saturated.	Selection clean. K ⁺ enrichment dominates the forward forcing cathodically.

Step	What changed	Verdict
A.2 grid	10×5 (k_{hyd}, λ) grid at V_{kin} .	10/10 rungs converge; mass-balance at machine precision; selectivity gap at V_{kin} is only +5 pp at $\lambda=1$, $k_{\text{hyd}}^{\text{route}} = 10^{-1}$.
Step 6 plumbing	Four-ablation audit (form-build residual, $\text{p}K_a$ -shift context, Picard Γ update, diagnostics).	All four pass. Side finding: K^+ Boltzmann pile-up $c_K \approx 291$ nondim at the OHP makes a 5% sentinel floor unreachable below $R_{\text{inj}} \leq 10$ — K^+ is <i>piled up</i> , not depleted, at the OHP under cathodic bias.
v10b calibration	Literature locks: $\Gamma_{\text{max}}=0.047$ (V10A chain tightened), $k_{\text{des}}=1.0$ (Eyring engineering choice $\Delta G \approx 0.80$ eV), $C_S=0.20$ F/m ² (Bohra-Koper-Choi). Sensitivity sweeps D7-D1 (C_S bracket $\{0.05, 0.10, 0.20, 0.30\}$) and D7-D4 ($\Gamma_{\text{max}} \times k_{\text{des}} \times k_{\text{hyd}}$ $3 \times 5 \times 2$ matrix).	All hard gates pass across sensitivity brackets.
Step 9 B.2	Densified $14 \times 10 = 140$ -rung (k_{hyd}, λ) grid at V_{kin} on v10b. Step 9.5 added warm_start_floor bisection so $\lambda = 0 \rightarrow 0.1$ at high k_{hyd} converges.	140/140 converge; mass-balance $\leq 5 \times 10^{-13}$; baseline reproduces v10b A.2 to $\leq 10^{-3}$ relative.
Step 10 Phase D fit	Single-parameter fit of Δ_β (carbon-vs-Cu offset added to Singh’s β -coefficient: $\beta^{\text{carbon}} = \beta^{\text{Cu}} + \Delta_\beta$) against deck $\text{K}_2\text{SO}_4/\text{pH} \in [3.5, 4.5]$ $\text{H}_2\text{O}_2\%$. Brent on loss $ \text{H}_2\text{O}_2^{\text{model}} - 50.95 \text{ pp} $.	OUTCOME_C_NON_IDENTIFIABLE_flagged (see §5).

5 Current findings (Phase D verdict + bridge runs)

Phase D verdict, 2026-05-12

Δ_β alone cannot match the deck. Two facts force the verdict.

(i) Stern σ -mapping is degenerate. Loss is flat at 15.629 pp² across *11 orders of magnitude* of Δ_β ($\Delta_\beta \in [-1.55 \times 10^7, 0]$ pm²). Mechanism: the local Stern σ at the OHP from our PNP+Stern solve is $\sim 10^{-7}$ counts/pm², so rescaling Δ_β by orders of magnitude barely moves $\Delta \text{p}K_a$ within its domain of influence.

(ii) Uniform +15.6 pp overshoot. Across every Δ_β the optimiser tested, model max $\text{H}_2\text{O}_2\%$ is exactly 66.58 pp. Deck $\text{K}_2\text{SO}_4/\text{pH}$ 4 mean is 50.95 pp. Δ_β can’t close ~ 16 pp of overshoot regardless of magnitude.

Phase E (cation-series predictive screen, Cs/Na/Li with Δ_β locked) must *not* launch on this fit.

Steric-coefficient discrepancy (surfaced during Phase D bridge runs). The Bikerman steric coefficient a_{nondim} is physical only for counterions in the current solver — O_2 , H_2O_2 , and (critically) H^+ are seeded with **A_DEFAULT=0.01**, which corresponds to a hard-sphere radius $r \approx 14.9$ Å, $\sim 150\times$ larger than physical (H_3O^+ Stokes radius is 2.8 Å, giving $a \approx 6.6 \times 10^{-5}$). Consequence: the Bikerman cap on local H^+ concentration is $c_{\text{max}} \approx 1/a \approx 100$ nondim ≈ 120 mol/m³ with the default, vs. $\sim 1.8 \times 10^4$ mol/m³ with the physical value. H^+ is therefore artificially *clamped* $150\times$ tighter than reality at the OHP under cathodic bias.

Bridge runs in flight (May 12+): four runs at deck-baseline config (V10B kinetics, $C_S = 0.20$, no hydrolysis, no K_w): two with legacy `A_DEFAULT`, two with physical $a_{O_2}/a_{H_2O_2}/a_{H^+}$. *Treat any plateau-set-by-Levich finding as suspect until these disambiguate.*

What we do believe robustly.

- Architecture is sound: dynamic $K^+ + \Gamma_{MOH} +$ finite-rate kinetics + no Stern Γ -correction converges cleanly across the full (k_{hyd}, λ) grid at the v10b production point.
- Mass balance / sign / gate-convergence *all pass* on every rung of every sweep.
- The Δ_β -alone fit *cannot* close the gap, which is independent of the steric discrepancy — the σ flatness alone is enough to falsify Δ_β -only as a knob.

6 Identified gaps

6.1 Model gaps

1. **Dynamic-species Bikerman radii.** H^+ (and to lesser extent O_2/H_2O_2) are clamped tighter than physical at the OHP. Bridge runs queued.
2. **Singh formula’s geometric assumption (sandwich) is from CO_2RR -on-Cu.** ORR-on-carbon has different adsorbates ($O_2/OOH/OH$) so the geometric chain that gives $r_{H,EI} < r_{M-O}$ is not strictly justified. We carry Cu values as priors; transferability is a load-bearing assumption.
3. **Open-reservoir desorption.** MOH^0 that desorbs vanishes from the model. A 5th NP species (bulk MOH) would close mass conservation; deferred.
4. **No Phase 6δ (alkaline-form ORR).** If cation hydrolysis only ever produces a plateau (not a peak followed by decay), parallel $R_{2e,alk} / R_{4e,alk}$ reactions are likely required for the post-peak decay.
5. **Constant Stern ε_S .** Booth-equation variable $\varepsilon_S(E)$ (Storey–Bazant, Bohra 2019) out of scope.
6. **No surface-charge contribution from Γ_{MOH} .** We assume MOH^0 is truly neutral. If MOH^δ carries a partial charge on carbon, the Gauss balance changes.

6.2 Data gaps (open external asks)

1. **Tafel slope analysis cation-pH-Li-K-Cs.xlsx.** Requested 2026-05-08; not delivered. K^+ pH 6.39 workaround (extracted from `0_1MK2SO4data8-15-19.xlsx`) gives Tafel slopes 270–310 mV/decade across three cycles, $R^2 = 0.996$. Multi-cation, multi-pH LSV data still needed to calibrate $K_0^{R_{4e}}$ and $\alpha_{R_{4e}}$ — currently 10^{-18} placeholder ratio and 0.5 placeholder.
2. **IrOx ring-pH probe raw data.** Behind Linsey 2025 deck slides 5–9 (the local-pH protocol for ORR). Without it the surface-pH-side validation of the model is purely against the deck’s slide-27 single-point summary; we have no V - or cation-resolved truth.
3. **MOH adsorbate coverage measurement** on sp^2 -carbon under K_2SO_4 ORR conditions (XPS / surface-IR / impedance). No peer-reviewed source currently reports this; our Γ_{max} is a hard-sphere monolayer derivation.
4. **Carbon-specific C_S .** We use the Bohra–Koper–Choi metal-electrode consensus 0.20 F/m². Direct CMK-3 measurement would lock C_S rather than treat it as a sensitivity-bracket parameter.
5. **Bohra 2019 EES paper** (Ruggiero ref 71) not present in `Articles/`. Methodologically closest PNP+BV+Stern reference for our regime.

7 Questions for the chemistry team

Grouped by topic. Bold-marked items are the highest-leverage.

Q1. Mechanism — is cation hydrolysis really the right primary mechanism?

Q1: Mechanism

Q1.1. In K_2SO_4 at pH 4–6 on CMK-3, do you think the *primary* cation-dependent buffer is the hydrolysis reaction $\text{K}(\text{H}_2\text{O})_n^+ \rightleftharpoons \text{K}(\text{H}_2\text{O})_{n-1}\text{OH}^0 + \text{H}^+$, or could something else (sulfate buffering, alkaline-form ORR, peroxide reduction) be load-bearing?

Q1.2. The deck's $\text{H}_2\text{O}_2\%$ curve has a cathodic peak that *decays* at more negative V_{RHE} . We can produce a plateau but not a decay. Mechanistically, what does your group think causes the decay? (Candidates: parallel alkaline-form $4e^-$ ORR turning on at low pH_{surf} , secondary peroxide reduction on CMK-3, mass-transport effects.)

Q1.3. Is $\text{Cu} \rightarrow \text{carbon}$ transfer of Singh's $\text{p}K_a$ -shift formula something the group has already independently tested or considered to fail? Singh's $r_{\text{H,El}}$ values depend on his adsorbed-CO sandwich geometry, which presumably doesn't hold on carbon.

Q1.4. Sulfate buffering: $\text{HSO}_4^-/\text{SO}_4^{2-}$ ($\text{p}K_a \approx 1.99$). At pH 4 the HSO_4^- fraction is small, so we've treated it as out of scope — is this consistent with what you observe, or do your CP curves show any sulfate-buffer signature?

Q2. Surface pH — what does the IrOx probe actually say?

Q2: Local pH measurement

Q2.1. Phase 6 α predicts surface pH ≈ 10.6 at $V_{\text{RHE}} = -0.40$ V on $\text{Cs}^+/\text{SO}_4^{2-}$ at $L_{\text{eff}} = 16$ μm with water self-ionization on. What does the IrOx ring-pH probe measure under matched conditions? Is the model way off, or just off by a few units?

Q2.2. Can we get the underlying IrOx data behind Linsey 2025 ACS-CATL deck slides 5–9 as a standalone file (raw ring currents + pH calibration)? Without it we're validating against the slide 27 single-point table.

Q2.3. Across the four cations (K, Cs, Na, Li) at matched j_{disk} , does the *shape* of pH_{surf} vs. V_{RHE} differ, or just the magnitude? Phase 6 β predicts cation-specific magnitudes via Singh; cation-specific shape would be evidence of more than just hydrolysis driving the difference.

Q3. Tafel + kinetic calibration data

Q3: Kinetic data

Q3.1. (highest leverage) Can we get the full Tafelslopeanalysiscation-pH-Li-K-Cs.xlsx, including the non-pH-6.39 LSV cycles? We have summary statistics for disks 2–6 (pH 5.21, 4.21, 3.42, 2.35, 1.65) but *no raw E-vs-j traces* in the workbook we currently hold. The full multi-pH data is the calibration source for $K_0^{R_{4e}}$ (currently a 10^{-18} placeholder ratio against $K_0^{R_{2e}}$) and $\alpha_{R_{4e}}$ (currently 0.5 placeholder).

Q3.2. Were the RDE/RRDE rotation rates logged with the LSV sweeps? We need this for the Koutecky–Levich correction — without it the ~ 280 mV/decade Tafel slope at K^+ /pH 6.39 is mass-transport contaminated and can't be cleanly compared to the kinetic-region textbook 60–120 mV/decade.

Q3.3. Is the four-cation chronopotentiometry data in the *_10-9-20.mat files (K, Cs, Na, Li) the canonical cation-series dataset, or is there a parallel LSV series elsewhere? For Phase E (the predictive screen) we'd compare predicted $\text{H}_2\text{O}_2\%$ to a measured cation-series curve.

Q4. Stern / OHP / capacitance

Q4: Stern capacitance and OHP geometry

Q4.1. Does anyone in your group or your collaborators (Bohra/Koper/Choi orbit) have a CMK-3-specific Stern capacitance measurement? We are currently using the metal-electrode Bohra–Koper–Choi consensus $C_S = 0.20$ F/m² as a prior. Carbon EDLC measurements at sp² are slightly lower (the carbon-conservative low end is ~ 0.05 F/m²).

Q4.2. For our purposes is it acceptable to treat C_S as field-averaged (constant)? Booth-equation variable $\varepsilon_S(E)$ is out of scope for v10b but could land if needed.

Q4.3. What's the right number for δ_{OHP} (the cation-OHP–electrode distance)? We use Bohra 2019's 0.40 nm; group-specific value would be cleaner.

Q4.4. Could a partially-charged MOH species (MOH^δ) on carbon contribute to Stern surface charge? Our v9 architecture assumes MOH is electrically neutral; if not, the Gauss balance at the cathode changes.

Q5. Hydrolysis kinetics — k_{des} , Γ_{max}

Q5: Hydrolysis rates and coverage

Q5.1. Γ_{max} is currently $\approx 5.6 \times 10^{-6}$ mol/m² (one hard-sphere K^+ monolayer at $r = 2.3$ Å). Is there any direct MOH adsorbate coverage measurement on sp²-carbon under K_2SO_4 ORR conditions (XPS, surface-IR, impedance) that we could use to anchor this? We searched and could not find one.

Q5.2. k_{des} (MOH^0 desorption rate from the OHP) is currently an engineering choice corresponding to $\Delta G \approx 0.80$ eV (Eyring at 298 K). Any experimental or DFT anchor for this?

Q5.3. Singh 2016 gives $K_{\text{eq}} = k_{\text{hyd}}/k_{\text{prot}}$ but not k_{hyd} or k_{prot} individually. Is there literature anchoring the forward hydrolysis rate? We can run the model at any k_{hyd} on the Langmuir-cap branch (coverage saturates at Γ_{max}), but the proton flux per area scales linearly with k_{hyd} up to saturation.

Q6. The selectivity-overshoot diagnosis

Q6: The +15.6 pp overshoot

Q6.1. (highest leverage) Model max $\text{H}_2\text{O}_2\%$ at the locked v10b production point is 66.58 pp *independent* of Δ_β across 11 orders of magnitude. The deck $\text{K}_2\text{SO}_4/\text{pH } 4$ mean is 50.95 pp. Two candidate root causes:

- (a) the H^+ Bickerman clamp (the $\sim 150\times$ -too-tight steric coefficient discussed above) is artificially capping H^+ at the OHP, which makes the $4e^-$ channel (whose rate $\propto c_H^4$) disproportionately suppressed and therefore selectivity towards $2e^-$ artificially high;
- (b) a different physics gap (e.g. alkaline-form $4e^-$ ORR turning on at low pH_{surf}) is missing entirely from the model.

Do you have a strong prior on which is the more plausible culprit?

Q6.2. On the deck, does the $\text{H}_2\text{O}_2\%$ peak position shift with pH? In K^+ between the six pH values (1.65 to 6.39), how much does the peak voltage / peak height change? This is a mechanism falsification test that doesn't require ring-pH probes.

Q6.3. Have you or Yash's group done a fit with explicit peroxide-reduction kinetics? Our solver currently has none — H_2O_2 is treated as an inert product. If real peroxide reduction is what produces the cathodic decay ($\text{H}_2\text{O}_2 + 2\text{H}^+ + 2e^- \rightarrow 2\text{H}_2\text{O}$ on CMK-3), that's missing entirely.

Q7. Process / next steps

Q7: Process

Q7.1. The Phase D fit *falsified* Δ_β alone. The candidate next-scope items (Phase D' / 6γ) are: (a) re-fit $k_{\text{des}} / \Gamma_{\text{max}}$ (currently literature-locked); (b) sweep $r_{\text{H,El}}$ as a calibration parameter (currently Cu-prior locked); (c) couple to local pH via mass-transport at the OHP; (d) revisit the Singh formula structure. Does your group have a preference about which of these we should chase first?

Q7.2. Is Phase 6δ (parallel alkaline-form ORR kinetics) something your group has already explored or has priors on? If so, k_0/α for $R_{2e,\text{alk}}$ and $R_{4e,\text{alk}}$ priors would shorten the search drastically.

Q7.3. What's the right falsification test for our model? "Predict the four-cation series with K^+ -calibrated parameters and check against the four-cation CP data" is the obvious one. Is there a smaller intermediate test (e.g. predict the K^+ Tafel slope vs. pH using one set of $k_{\text{hyd}}/k_{\text{des}}$) that you'd find more diagnostic?

Pointers

- Phase 6α outcome: docs/phase6/PHASE_6A_INVESTIGATION_SUMMARY.md.
- Phase 6β live plan: docs/phase6/phase6b_next_steps_plan.md.
- Singh 2016 extraction: docs/phase6/singh_2016_pka_formula.md.
- v10b calibration: docs/phase6/v10b_calibration_summary.md.
- Step 9 B.2 (140-rung grid): docs/phase6/phase6b_step9_B2_summary.md.
- Step 10 Phase D verdict: docs/phase6/phase6b_step10_phase_D_summary.md.
- Cs^+ vs. K^+ audit: docs/phase6/CONJECTURE_AUDIT_2026-05-09.md.
- Open data asks: docs/phase6/missing_data.md.

- Source-of-truth solver code: `Forward/bv_solver/water_ionization.py`, `Forward/bv_solver/cation_hydrolysis.py`.
- Companion long-form writeup: `writeups/May13th/phase_6_overview.tex`.